Introduction to Compressed Sensing. Notions of measurements and sparsity

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Abstract

Compressed Sensing (CS), or compressed acquisition, is a data-processing problem in which the acquisition and compression steps are carried out at the same time, during acquisition, by an "intelligent" measurement procedure. Such a procedure yields large gains in storage performance as well as in the cost and speed of acquisition. A great many real data sets can be rewritten in the CS paradigm. In this part we present several concrete examples in which CS is a relevant model.

A key point in the success of CS is, of course, that the data stored or observed after acquisition must be able to recover the original signal, by means of fast, stable, and robust algorithms. We shall see that this reconstruction can be performed efficiently for certain types of signals that admit a "sparse" representation in certain bases.

The notion of sparsity is central to the theory of CS. The aim of this first part is to introduce this notion and to show that it arises naturally in real data.

In this first part we therefore introduce two key notions in CS: measurements and sparsity. The formalization and modeling of the CS problem, and of several related problems, are also given.

1 Introduction

Compressed sensing may be rendered as "compressed acquisition". This means that the acquisition and the compression of a signal are performed in a single step.

Before CS, data compression was a step separate from acquisition. In digital photography, for example, the first step is to take the picture, that is, to expose the camera's photosensitive plate to the light of the scene to be photographed. These data are stored in RAW format: the scene is pixelized and the color level of each pixel is stored in a large matrix. For present-day cameras, storing a photograph in RAW format occupies on the order of a megabyte of memory. This is the first step: that of **acquisition**. The second step is generally carried out by a CPU (either internally in the camera, or on a PC after the RAW files have been transferred). It consists in running a compression algorithm on the RAW file associated with the image. This second step is the **compression** of the signal, which makes it possible to divide the storage space by up to a factor of 25, that is, about a hundred kilobytes of storage for a photograph in compressed format instead of a few megabytes in RAW format. The JPEG format is a classical extension for photographs compressed by a certain algorithm based on transforming the images (by blocks) in a cosine basis and on a quantization of the resulting coefficients. Compression is useful only if one can reconstruct the original image from the compressed one quickly and efficiently: that is, without too much loss—roughly speaking, the human eye should not see the difference between the original image and the image reconstructed after compression and decompression. This is

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the case for JPEG compression, which, despite a loss of information due (among other things) to the quantization step for the coefficients of the DCT (discrete cosine transform), permits a reconstruction of images that is very faithful to the originals.

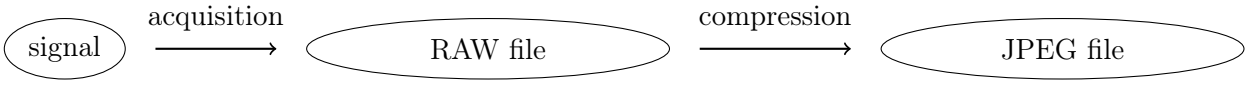


Figure 1: The two steps in the classical approach: 1) acquisition 2) compression.

Unlike the classical approach, CS proposes to carry out the two preceding steps of acquisition and compression in a single step; hence the term compressed acquisition. The aim is therefore to take "intelligent" measurements of the signal one wishes to know (or to estimate), in such a way that this signal can be reconstructed from only a small number of these measurements.

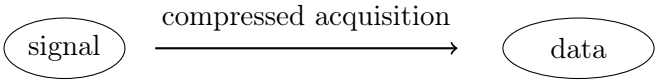


Figure 2: In CS, acquisition and compression are performed in one and the same step.

The notion of a **measurement** is therefore central in CS. A large part of the theoretical work that has taken place (and that is still continuing) in research on CS concerns the construction of measurements that permit compressed acquisition. In CS, one considers only linear measurements of signals. If x is the signal we wish to know, one then takes measurements of the form

$$y_i = \langle x, X_i \rangle \tag{1.1}$$

where:

1. i: the index of the measurement,
2. X_i: the i-th measurement vector,
3. y_i: the i-th measurement, or observation.

We denote by m the number of measurements. We shall be interested only in the reconstruction of signals in finite-dimensional spaces. We therefore assume that x belongs to $\mathbb{R}^N$ (or $\mathbb{C}^N$—to simplify the presentation, we consider only the case of $\mathbb{R}^N$ here). One may then rewrite the observed measurements in matrix form:

$$y = Ax \tag{1.2}$$

where:

1. $y \in \mathbb{R}^m$: the vector of measurements,
2. $A \in \mathbb{R}^{m \times N}$: the measurement matrix, whose rows are the measurement vectors $X_i^\top$,

$$A = \begin{pmatrix} X_1^\top \\ \vdots \\ X_m^\top \end{pmatrix} \tag{1.3}$$

3. $x \in \mathbb{R}^N$: the signal to be reconstructed.

The aim being to take as few measurements as possible—in order to minimize the storage cost or the cost of acquisition—one will always have

$$m < N \tag{1.4}$$

that is, the number m of measurements will in general be (much) smaller than the dimension N of the ambient space (the space in which the signals to be reconstructed live). This is the classical setting of **high-dimensional statistics**: the number m of observations is smaller than the dimension N of the space of parameters to be estimated (the signal x is viewed as the parameter indexing the statistical model).

For the moment, the problem in CS is stated as follows: recover $x \in \mathbb{R}^N$ from y $(= Ax)$ and from A. From an algebraic point of view, one is therefore led to solve a linear system with m equations and N unknowns (the coordinates of x), while $m < N$ (and even, in general, $m << N$). Figure 3 represents the problem as a "highly underdetermined" problem: there are many more unknowns than equations. There are therefore several solutions $x \in \mathbb{R}^N$ such that $y = Ax$, and hence, without an additional assumption on x, one will not be able to reconstruct x from y and A as soon as $m < N$.

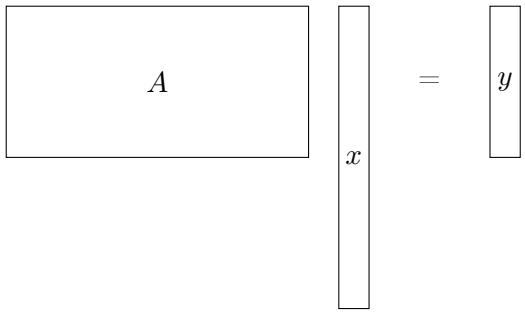


Figure 3: Finding x from $y = Ax$ and A: Compressed Sensing may be viewed as the problem of solving a highly underdetermined linear system.

The central assumption in CS is that the signal x to be reconstructed is **sparse**—that is, most of the coordinates of x are zero. Equivalently, the size of the support of x (the set of indices of the nonzero coordinates of x) is small compared with N.

Definition 1.1 *Let N and s be two integers such that $1 \leq s \leq N$. Let $x \in \mathbb{R}^N$. The **support of** x is the set*

$$\mathrm{supp}(x) = \big\{j \in \{1, \ldots, N\} : x_j \neq 0\big\},$$

where, for every $1 \leq j \leq N$, x_j is the j-th coordinate of x. The size of the support of x is denoted

$$\|x\|_0 = |\mathrm{supp}(x)|,$$

*and is the number of nonzero coordinates in x. One says that a vector is **s-sparse** (or sparse) when $\|x\|_0 \leq s$, and one denotes by Σ_s the set of all s-sparse vectors in $\mathbb{R}^N$:*

$$\Sigma_s = \big\{x \in \mathbb{R}^N : \|x\|_0 \leq s\big\}.$$

We therefore assume that the signal x is s-sparse. We shall later see examples in which this assumption is well satisfied by "concrete" signals.

Remark 1.2 *In Definition 1.1, the canonical basis plays a central role: the notion of sparsity is defined with respect to the canonical basis. This definition extends to any basis, or even to dictionaries (sets of signals with no particular structure). If $\mathcal{F} = \{f_1, \cdots, f_p\}$ is a set of elements of $\mathbb{R}^N$ and $x \in \mathbb{R}^N$, one says that x is s-sparse in the dictionary $\mathcal{F}$ when there exists $J \subset \{1, \ldots, p\}$ such that*

$$|J| \leq s \text{ and } x = \sum_{j \in J} \theta_j f_j.$$

Thus x has a concise expression in $\mathcal{F}$: as a linear combination of only s elements of $\mathcal{F}$. One then speaks of ***structured sparsity****.*

Finding the "good bases" or "good dictionaries" in which certain types of signals (images, video, and so on) have sparse representations has been (and still is) the subject of a great deal of research in approximation theory. For example, it turns out that certain wavelet bases are, for many types of "real" data, good bases from this point of view. This aspect of signal processing will not be studied in this course (we refer the interested reader to the book [6] or to the numerical tours of [7]). In what follows we shall therefore consider only the notion of unstructured sparsity, i.e., the notion defined with respect to the canonical basis as in Definition 1.1. The general case reduces to the following transformation, $x = F\theta$, where $F \in \mathbb{R}^{N \times p}$ is the matrix whose columns are given by the f_j, $j = 1, \ldots, p$, and $\theta \in \mathbb{R}^p$ is an s-sparse vector, and hence to replacing the measurement matrix A by AF. The sparsity of x in $\mathcal{F}$ is therefore expressed by the sparsity of θ in the canonical basis.

Remark 1.3 *The three weak points of the two-step "acquire-then-compress" approach are:*

1. *The acquisition phase requires the storage of N data, whereas for the problems of interest to us N is typically large (a storage problem).*
2. *One must compute all N coordinates of the signal in the compression basis, whereas only a small proportion of them will be kept (a problem of computation time used unnecessarily).*
3. *The locations of the large coefficients in the compression basis must be stored for each image.*

In what follows we present several concrete problems in which the data can be translated into the CS framework. We begin, however, by giving in the next section a formalization of the CS problem and two important variants (noisy data and an "approximately sparse" signal).

2 Formalization of the compressed sensing problem

In the previous section, the notions of *measurement* and of *sparsity* were introduced. Sparsity is a fundamental assumption on the signals to be reconstructed, and the measurements are in some cases freely chosen by the statistician and in other cases "imposed" by nature. In the first case, in which the statistician chooses the measurements, that is, the measurement vectors X_i introduced in (1.1), one may speak of an experimental design and hence of a certain strategy for choosing the measurements. The aim is to be able to reconstruct the (sparse) signal from a minimal number of measurements. In this case, one may then state the CS problem as follows.

CS problem number 1 : Construct, in minimal number, measurement vectors $X_1, \ldots, X_m$ such that for every vector $x \in \Sigma_s$, one can reconstruct x "efficiently" from y $(= Ax)$ and from $A \in \mathbb{R}^{m \times N}$, the measurement matrix defined in (1.3)

It remains to define what is meant by "efficiently". Here, and for the moment, an efficient reconstruction of x is valid when one can implement an algorithm that actually works, i.e., that does not require too much time or too much computational power. Later we shall give a more precise definition of the algorithms one refrains from implementing in practice (NP-hard algorithms) and of those that are particularly valued (linear programming or semidefinite programming). There is therefore a very important "computational" component in CS. We shall also address it in this course.

CS problem number 2 : Given measurement vectors $X_1, \dots, X_m$, construct an efficient reconstruction algorithm for every vector $x \in \Sigma_s$, from y $(= Ax)$ and from $A \in \mathbb{R}^{m \times N}$ (defined in (1.3)). That is, a map

$$\Delta : \mathbb{R}^m \mapsto \mathbb{R}^N \tag{2.1}$$

that can be implemented efficiently and such that

$$\text{for all } x \in \Sigma_s, \Delta(Ax) = x. \tag{2.2}$$

(there is a slight abuse of notation here, because the reconstruction algorithm Δ uses A; one should therefore write $\Delta(Ax, A)$ instead of $\Delta(Ax)$).

We shall return to the question of the efficiency of algorithms in the later sections concerning ℓ_0 and ℓ_1 minimization algorithms. For the moment, the important point is that the measurements are to be chosen so as to need as few measurements as possible in order to reconstruct (efficiently) any s-sparse signal x. We stress that one wishes to be able to identify m measurement vectors $X_1, \dots, X_m$ that make it possible to reconstruct (efficiently) all vectors $x \in \Sigma_s$ from y $(= Ax)$ and A alone, where A is the same measurement matrix.

One will therefore use the same set of measurement vectors $\{X_1, \dots, X_m\}$ for all s-sparse signals x. This notion of uniformity over Σ_s plays an important role in determining the minimal number m of measurements that permits efficient reconstruction over Σ_s.

There are therefore mainly two problems in CS:

1. a problem of choosing the measurement vectors $X_1, \dots, X_m$ in minimal number,
2. a problem of constructing algorithms for the reconstruction of s-sparse signals from a highly underdetermined system of equations: $\Delta : \mathbb{R}^m \mapsto \mathbb{R}^N$ such that for every $x \in \Sigma_s$, $\Delta(Ax) = x$.

These two problems are often treated at the same time; that is, one constructs measurement vectors so that a given algorithm satisfies (2.2).

Remark 2.1 *Compressed sensing is an example of a high-dimensional statistics problem, because the number m of observations—here called measurements—is much smaller than the dimension of the object one wishes to estimate, here N. Intuitively, the "true" dimension of an s-sparse vector is s and not N—this would genuinely be the case if one knew the support of x. In the case of an estimation problem for an object of dimension s, the minimal number of observations that must be obtained for the estimation to be nontrivial is at least s. One would therefore like, for the CS problem, the minimal number of measurements sufficient to reconstruct every s-sparse vector to be likewise of order s, to which one should, in some cases, add a cost because the support of the signal is unknown. Intuitively, one will work in the regime $s << N$ and, "ideally", $m \sim s$—this gives an idea of the order of magnitude expected for the number of measurements.*

In the two subsections that follow, we introduce two "more realistic" variants of the compressed sensing problem introduced above.

2.1 Noisy data – robust procedures

In many situations, the observation of the data is noisy. One then observes linear measurements of a signal x to which a noise has been added. These data are modeled by:

$$y_i = \langle X_i, x \rangle + \xi_i \text{ for all } i = 1, \dots, m \tag{2.3}$$

where ξ_i is the noise term on the i-th observation. In statistics, it is standard to model the noises $\xi_1, \dots, \xi_m$ as a family of m random variables, generally independent and identically distributed, and in many cases this distribution is centered Gaussian with (unknown) variance σ. In matrix notation, one observes

$$y = Ax + \xi \tag{2.4}$$

where $y = (y_i)_{i=1}^m$, A is the measurement matrix defined in (1.3), and $\xi = (\xi_i)_{i=1}^m$ is the noise vector.

In the case of noisy data, it is no longer possible to reconstruct x exactly from $y = (y_i)_{i=1}^m$ and A as in (2.2). One will therefore seek to estimate (still efficiently—in a *computational* sense) the signal x. One will wish this reconstruction to be robust in the sense of the following definition.

Definition 2.2 *One says that a reconstruction (or decompression) algorithm $\Delta : \mathbb{R}^m \mapsto \mathbb{R}^N$ is* ***robust (to noise) of order*** **s** *when, for every $x \in \Sigma_s$ and every noise vector $\xi \in \mathbb{R}^m$,*

$$\|x - \Delta(Ax + \xi)\|_2 \leq c_0 \|\xi\|_2$$

where c_0 is an absolute constant and $\|t\|_2 = \left(\sum_{i=1}^m t_i^2\right)^{1/2}$ for every $t = (t_i)_{i=1}^m \in \mathbb{R}^m$.

The idea behind the robustness property is that if there is no noise—i.e., $\xi = 0$—then Δ is indeed an exact reconstruction algorithm on Σ_s, because $\Delta(Ax) = x$ for every $x \in \Sigma$. But if the data are noisy, one still wishes to ensure that the reconstruction of x by $\Delta(Ax+\xi)$ makes an error of at most the order of magnitude of the noise, measured here by the ℓ_2^m norm.

For many concrete problems, robustness to noise is a necessity, because even if the data are noisy one may hope that the magnitude of the noise (measured here by $\|\xi\|_2$) will not be too large relative to the amplitude of the signal (here $\|x\|_2$). In statistics, $\|x\|_2 / \|\xi\|_2$ is called the SNR—*signal-to-noise ratio*. The SNR measures the feasibility of the denoising problem: when the SNR is small, the signal is "drowned" in the noise, and one therefore has very little hope of estimating x well, and conversely.

2.2 Approximately sparse signals – stable procedures

The signals one seeks to reconstruct are not, in general, exactly s-sparse, but only "close" to an s-sparse signal. One would therefore like to ensure that the procedures we have constructed (under the belief that the signals to be estimated are exactly s-sparse) behave well in the face of this possibility. This property is called *stability*.

Definition 2.3 *One says that a reconstruction (or decompression) algorithm $\Delta : \mathbb{R}^m \mapsto \mathbb{R}^N$ is* ***stable of order*** **s** *when, for every $x \in \mathbb{R}^N$,*

$$\|x - \Delta(Ax)\|_2 \leq c_0 \frac{\min_{z \in \Sigma_s} \|x - z\|_1}{\sqrt{s}}$$

where c_0 is an absolute constant and $\|t\|_1 = \sum_{i=1}^m |t_i|$ for every $t = (t_i)_{i=1}^m \in \mathbb{R}^m$.

The choice of norms in the stability property is important. One sees that the reconstruction error "$\|x - \Delta(Ax)\|_2$" is measured with respect to the ℓ_2^N norm, whereas the approximation error "$\min_{z\in\Sigma_s} \|x - z\|_1$" is measured with respect to the ℓ_1^N norm. Ideally, one would have liked the approximation error to be measured in the ℓ_2^N norm as well; however, one can show (we shall see this in an exercise) that if there exists a reconstruction algorithm Δ such that for every $x \in \Sigma_s$, even for $s = 1$, satisfying $\|x - \Delta(Ax)\|_2 \leq c_0 \min_{z\in\Sigma_s} \|x - z\|_2$, then necessarily $m \geq c_1 N$, where c_1 is an absolute constant depending only on c_0. One therefore cannot have the stability property for an approximation error measured in the ℓ_2^N norm unless the number of measurements is of the order of magnitude of the dimension of the space—which one wishes to avoid at all costs.

It is easy to see that the approximation term satisfies:

$$\sigma_s(x)_1 := \min_{z\in\Sigma_s} \|x - z\|_1 = \|x - x_s^*\|_1$$

where x_s^* is the vector in Σ_s consisting of the s largest coordinates of x in absolute value, that is, such that for every $1 \leq j \leq N$,

$$(x_s^*)_j = \begin{cases} x_j & \text{when } j \in J_s \\ 0 & \text{otherwise} \end{cases}$$

where $J_s = \{j_1, \ldots, j_s\}$ and $|x_{j_1}| \geq |x_{j_2}| \geq \cdots \geq |x_{j_s}| \geq x_j$ for every $j \notin J_s$. Thus the s-sparse vector closest to x (for the ℓ_1^N norm) is simply a truncated vector in which only the s largest coordinates of x in absolute value have been kept.

The stability property of an algorithm ensures that if x is indeed s-sparse, then the reconstruction is indeed exact (i.e., $\Delta(Ax) = x$ for every $x \in \Sigma_s$), but if the signal x is not exactly s-sparse and is only close to an s-sparse vector, then $\sigma_s(x)_1$ will be small, and in that case the reconstruction $\Delta(Ax)$ will not be far from x when Δ satisfies the stability property. As in the case of robustness, the estimation error is of the order of the error relative to the "ideal" model (no noise, and an exactly sparse signal). Here the ideal setting is not satisfied, because the signal is not exactly s-sparse, and one therefore pays a price measured by this approximation error $\sigma_s(x)_1$, since our procedures will have been designed and constructed in the ideal setting, even though we shall adapt them to the more realistic cases of noisy data and of signals that are not exactly s-sparse.

Finally, the only algorithms that will be of practical interest must combine the two properties of robustness and stability. Indeed, one frequently encounters, on real data, the presence of noise in the measurements and of approximately sparse signals. The following definition brings these two aspects together.

Definition 2.4 *One says that a reconstruction (or decompression) algorithm $\Delta : \mathbb{R}^m \mapsto \mathbb{R}^N$ is* ***stable and robust of order*** **s** *when, for every $x \in \mathbb{R}^N$ and every $\xi \in \mathbb{R}^m$,*

$$\|x - \Delta(Ax + \xi)\|_2 \leq c_0 \frac{\min_{z\in\Sigma_s} \|x - z\|_1}{\sqrt{s}} + c_0 \|\xi\|_2$$

where c_0 is an absolute constant.

One session of the course will be devoted to the study of algorithms satisfying these two properties. The remainder of these introductory lecture notes on CS concerns concrete examples. The presentation of these examples will place particular emphasis on the central notions of (linear) measurements and of sparsity.

Remark 2.5 *Up to now, we have considered measurements of the form $y = Ax + \xi$, where A is the $m \times N$ matrix whose m row vectors are given by the measurement vectors X_i, $1 \leq i \leq m$. It is, however, common to encounter another normalization, in which the measurement matrix is normalized by $1/\sqrt{m}$ (this normalization is of interest in probability because it brings out empirical means). The data are then*

$$y' = A'x + \xi'$$

where $y' = y/\sqrt{m}$, $A' = A/\sqrt{m}$, and $\xi' = \xi/\sqrt{m}$. For this choice of normalization, a procedure is stable and robust of order s when, for every $x \in \mathbb{R}^N$ and every $\xi' \in \mathbb{R}^m$,

$$\left\|x - \Delta(A'x + \xi')\right\|_2 \leq c_0 \frac{\min_{z \in \Sigma_s} \|x - z\|_1}{\sqrt{s}} + c_0 \frac{\|\xi'\|_2}{\sqrt{m}} \tag{2.5}$$

where c_0 is an absolute constant.

This is the normalization customarily used in statistics. In this setting, it is standard to assume that the noise ξ' is a standard Gaussian vector in $\mathbb{R}^m$. For this model, $\|\xi'\|_2$ is, with very high probability (at least $1 - 2\exp(-c_0 m)$), of the order of magnitude of $\sqrt{m}$, so the term $\|\xi'\|_2 / \sqrt{m}$ is constant in (2.5). This is optimal when $m \sim s \log(eN/s)$.

When one has more observations than necessary (that is, more than $s \log(eN/s)$), one can construct procedures $\widehat{x}$ (Lasso or Dantzig) that satisfy, with high probability,

$$\|x - \widehat{x}\|_2 \lesssim \inf_{z \in \Sigma_s} \Big(\|x - z\|_2 + \sqrt{\frac{\|z\|_0 \log(n)}{m}} \Big).$$

But this result holds only for a single signal $x \in \mathbb{R}^N$ to be reconstructed, whereas Definition 2.4 requires that the result hold for every vector $x \in \Sigma_s$.

3 Compressed sensing for photography – example of the single-pixel camera

The example that served as our running thread in introducing the notion of compressed acquisition "in a single step" in the first section was that of taking photographs. We continue the presentation of this example in this section.

The cameras we currently use are based on the older paradigm in data compression, which operates in two steps: 1) acquisition 2) compression (cf. Figure 1). What would a camera look like that uses the central idea of CS, that of compressed acquisition (in a single step)?

The aim is therefore to construct measurement vectors—in small number—such that the stored data make it possible to reconstruct the photographed scene. Instead of recording the color level for each pixel of the camera's photosensitive plate, one takes a linear measurement of these color levels. This amounts to taking a local average of the color level according to the resolution and the brightness available, and then taking a weighted sum of all these color levels. Figure 4 shows an example of a measurement of a scene.



Figure 4: Linear measurement of a scene. Here the vector x one wishes to reconstruct is a locally averaged scene (each pixel represents the average white level of a real scene)—this is the image on the left (the "classical" photograph of Lena). It is therefore a matrix here. The measurement X_i is the image on the right, where the level of black indicates the weight assigned to the pixel with which it is associated.

In the classical approach, the acquisition step amounts to taking measurements as shown in Figure 4, except that the measurement vector equals 0 everywhere except at one pixel, where it equals 1. There are therefore as many measurements as pixels—hence the large size of RAW files.

For the construction of a camera using the ideas of compressed acquisition, one will take several measurements of the same "type" as those in Figure 4. One will therefore construct several measurement vectors and record the inner product between the observed scene and the measurement vector. In Figure 5, one stores the measurements $y_1, \ldots, y_m$. The aim is to be able to reconstruct the original scene (pixelized at the given resolution) from the measurements y_i.



Figure 5: m measurements of a scene.

It is in particular interesting to observe that the values $y_1, \ldots, y_n$ are linear combinations of color levels (or of black and white); they are therefore real values that can be recorded with a photosensitive plate containing only a single photosensitive site (in contrast to the photosensitive plates currently in use, which contain several million photosensitive sites). This observation is at the origin of the construction, by Rice University, of the "single-pixel camera".

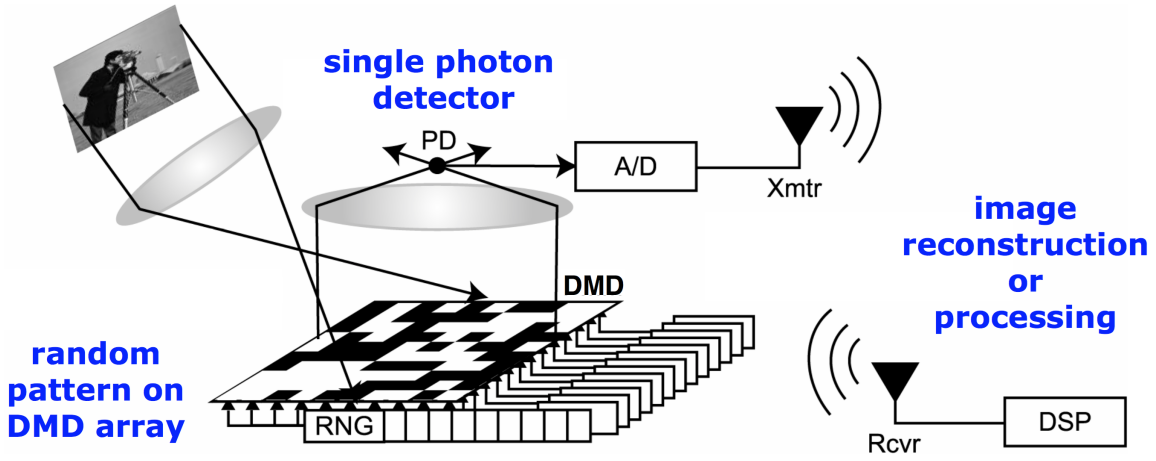


Figure 6: Single-pixel camera of Rice University. The measurement vector is constructed by an array of DMDs (*digital micromirror device*) that orient themselves at random. The measurement y_i is collected by a single photon detector.

The construction of the measurement vectors in the single-pixel camera is carried out by an array of micromirrors which, by orienting themselves, let more or less light through. It is the sum of all these more or less attenuated light levels that is then sent to the single photosensitive detector. A key point in the construction of the measurement vectors is the presence of randomness.

We shall indeed see later that an efficient construction of the measurement vectors proceeds (at the current state of knowledge) by constructions of random measurements. This is therefore the strategy used by Rice University for the choice of orientation of all the micromirrors. Figure 7 offers a close-up of these micromirrors.

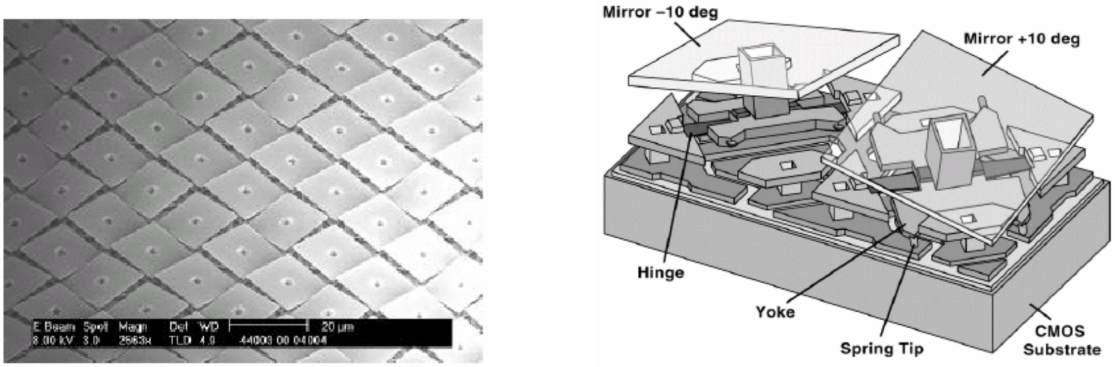


Figure 7: Close-up of the micromirrors used in the single-pixel camera of Rice University. The orientation, and hence the level of opacity, of these mirrors is obtained at random for the construction of measurement vectors.

It therefore seems possible to build a camera using the idea of compressed acquisition in a single step. The problem now is to know whether it works. For this, a key assumption in CS is the sparsity assumption: however complicated the signal may be here, can it be described by a low-dimensional structure? This is the typical question one asks in CS, and more generally in high-dimensional statistics. Here, one asks whether images can be described concisely—i.e., with only a small number of real numbers.

This question is much older than the CS problem. Many results were obtained in the 1980s and 1990s, notably thanks to the introduction of wavelets. The central idea of wavelets is to propose bases that are localized both in time and in frequency (in contrast to the Fourier basis, which is localized only in frequency). Engineers and researchers in signal processing have succeeded in constructing (orthonormal) bases in which the expansion coefficients of "real" photographs are, in the great majority, very small relative to very large coefficients that are present in small number.

Formally, if one represents an image as an array of numbers $f \in \mathbb{R}^N$—where N is here of the form $N = N_1 \times N_2$ (N_1 being the number of rows and N_2 the number of columns)—then one can construct orthonormal bases $(\psi_j)_{j=1}^N$ of $\mathbb{R}^N$ such that if one expands f in these bases,

$$f = \sum_{j=1}^{N} \langle f, \psi_j \rangle \psi_j$$

then the coefficients $(\langle f, \psi_j \rangle)_{j=1}^N$ of the expansion are in the great majority small (relative to other coefficients that are larger and that carry almost all of the interesting information, even though there are only a small number of them). So much so that if one keeps only the s largest coefficients in the expansion of f,

$$f_s^* = \sum_{j \in J_s} \langle f, \psi_j \rangle \psi_j$$

(where $J_s \subset \{1, \cdots, N\}$ is the set of indices of the s largest coordinates of $(\langle f, \psi_j \rangle : j = 1, \ldots, N)$), then the image rendered by f will in general be indistinguishable, to the human eye, from the one associated with f_s^* if one takes s of the order of 10 KB for an image of size $N \sim 1$ MB (a good illustration of this phenomenon can be found, for example, in Figure 1.1 of the book [4], where only 1% of the largest wavelet coefficients are kept between the original and the compressed image, which are nevertheless of the same quality to the human eye). Thus, surprising as this may seem (given the a priori complexity and diversity of photographs), photographs can be well approximated by objects of "small" dimension. The sparsity assumption is therefore "approximately" satisfied. If there exist procedures that are stable in the sense of Definition 2.3, then one will be able to reconstruct the images obtained by the single-pixel camera. Since the measurements will very likely be noisy, one will also require this algorithm to be robust to noise.

The bases used for image processing that permit a sparse representation of images are wavelet bases. In a few words, and to put it simply, they are constructed from a mother function, for example the Haar function, and from a scaling function. We begin by giving the construction of the Haar basis on $L^2([0,1])$, in order to present the idea of localization in time and in frequency of this basis, mentioned above. We shall then construct this basis on $L^2(\{0, 1, \cdots, n-1\})$ (i.e., for vectors in $\mathbb{R}^n$), because it will serve us, by tensorization, to construct the basis on $L^2(\{0, 1, \cdots, n-1\}^2)$ (i.e., for matrices of size $n \times n$), which is the one that really interests us for the (quasi-)sparse representation of photographs.

Construction of the Haar basis on $L^2([0,1])$. Here we wish to construct the (orthonormal) Haar basis for square-integrable functions from $[0,1]$ to $\mathbb{R}$. We begin by introducing the Haar mother function

$$\Psi : t \in \mathbb{R} \to \begin{cases} 1 & \text{if } 0 \leq t < 1/2 \\ -1 & \text{if } 1/2 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

and the scaling function, here the constant function $\Phi : t \in [0,1] \to 1$. The functions of the Haar basis are then obtained by translations and dilations of the mother wavelet (see Figure 8): for

every *resolution level* $j \in \mathbb{N}$ and every $i = 0, 1, \cdots, 2^j - 1$, one sets

$$\Psi_i^j : t \in [0,1] \to \sqrt{2^j}\Psi\left(2^j\left(t - \frac{i}{2^j}\right)\right).$$

The Haar basis of $L^2([0,1])$ is $(\Phi, \Psi_i^j : j \in \mathbb{N}, i = 0, 1, \cdots, 2^j - 1)$.

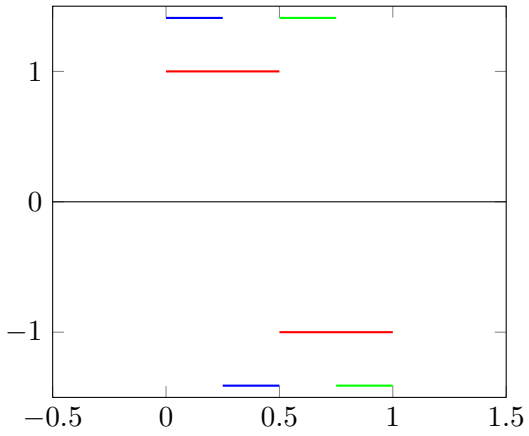


Figure 8: Mother function of the Haar basis in red. The two functions (for $i = 0, 1$) of the Haar basis at resolution level $j = 1$, in blue and green.

One may ask what the interest of the Haar basis is, compared with the (better known) Fourier basis $(F_j)_{j\in\mathbb{N}}$, where $F_0 : t \in [0,1] \to 1$ and, for every $j \in \mathbb{N}^*$,

$$F_j : t \in [0,1] \to \begin{cases} 2\cos\left(\frac{2\pi jt}{2}\right) & \text{if } j \in 2\mathbb{N}^* \\ 2\sin\left(\frac{2\pi(j+1)t}{2}\right) & \text{if } j \in 2\mathbb{N}+1 \end{cases}$$

For example, if $f : t \in [0,1] \to \cos(2\pi t)$, then its Fourier coefficients are given by $\langle f, F_2\rangle = 1$ and $\langle f, F_j\rangle = 0$ for every $j \neq 2$. They are therefore all zero except the second. If f undergoes a small perturbation g_ϵ localized around $1/2$, then in general one will have $\langle f + g_\epsilon, F_j\rangle \neq 0$ for every $j \in \mathbb{N}$. Thus even a very small perturbation localized in time (here, locally around $1/2$) will have consequences on the entire spectrum (i.e., at every frequency level) of f: all the Fourier coefficients of f are modified by a small perturbation localized in time. This is not the case for bases localized in time and in frequency: a small perturbation localized in time will imply a perturbation localized in frequency. Here one will have $\langle f, \Psi_i^j\rangle = \langle f, \Psi_i^j\rangle$ at each frequency level j, except for a small number of indices i. In other words, the Haar basis is robust to perturbations localized in time, whereas the Fourier basis is not.

Construction of the Haar basis on $L^2(\{0, 1, \cdots, n-1\})$. The construction of a basis on $L^2(\{0, 1, \cdots, n-1\})$ is equivalent to the construction of a basis on $\mathbb{R}^n$ when one identifies a vector $f = (f_t)_{t=0}^{n-1} \in \mathbb{R}^n$ with a function $f : t \in \{0, 1, \ldots, n-1\} \to f_t \in \mathbb{R}$. The construction of a Haar basis for vectors in $\mathbb{R}^n$ is therefore easily deduced from the construction of a Haar basis for $L^2(\{0, 1, \cdots, n-1\})$ according to this vector/function correspondence.

For the construction of the Haar basis on $L^2(\{0, 1, \cdots, n-1\})$, one draws inspiration from the one obtained on $L^2([0,1])$. One defines a mother wavelet on $\mathbb{Z}$ by

$$\psi : t \in \mathbb{Z} \to \begin{cases} 1 & \text{if } 0 \leq t < (n-1)/2 \\ -1 & \text{if } (n-1)/2 < t \leq n-1 \\ 0 & \text{otherwise} \end{cases}$$

and the scaling function, here the constant function $\phi : t \in \{0, \cdots, n-1\} \to 1$. One takes n of the form $n = 2^J$ for some integer $J \in \mathbb{N}$. The functions of the Haar basis at resolution level $j = 0, 1, \cdots, \log_2 n - 1$ are then obtained by translations and dilations of the mother wavelet by

$$\psi_i^j : t \in \{0, 1, 2, \cdots, n-1\} = \sqrt{2^j}\psi\left(2^j\left(t - \frac{(n-1)i}{2^j}\right)\right)$$

for every $i = 0, 1, \cdots, 2^j - 1$. The Haar basis of $L^2(\{0, 1, \cdots, n-1\})$ is then $(\phi, \psi_i^j, j = 0, 1, \cdots, \log_2 n - 1, i = 0, 1, \cdots, 2^j - 1)$. One may also reorder this basis in order to have a more concise notation, by setting

$$w_0 : t \in \{0, 1, \cdots, n-1\} \to 1 \text{ and } w_k : t \in \{0, 1, \cdots, n-1\} \to \psi_i^j(t) \text{ for } k = 2^j + i \in \{1, \ldots, n-1\}.$$

Thus $(w_k)_{k=0}^{n-1}$ is the Haar basis of $L^2(\{0, 1, \cdots, n-1\})$ and hence also of $\mathbb{R}^n$.

Construction of the Haar basis on $L^2(\{0, 1, \cdots, n-1\}^2)$. It is the Haar basis on $L^2(\{0, 1, \cdots, n-1\}^2)$ that is useful for the sparse expansion of photographs when $n = N_1 = N_2$. It is obtained by tensorization of the basis constructed above on $L^2(\{0, \ldots, n-1\})$. For every $k_1, k_2 \in \{0, 1, \cdots, n-1\}$ and every $t_1, t_2 \in \{0, 1, \ldots, n-1\}$,

$$(w_{k_1} \otimes w_{k_2})(t_1, t_2) = w_{k_1}(t_1) w_{k_2}(t_2).$$

The function $w_{k_1} \otimes w_{k_2}$ defined on $\{0, 1, \ldots, n-1\}^2$ is called the tensor product of w_{k_1} and w_{k_2}. The Haar basis on $L^2(\{0, \ldots, n-1\}^2)$ is given by $(w_{k_1} \otimes w_{k_2} : k_1, k_2 \in \{0, \ldots, n-1\})$. Thus, the wavelet transform of a signal $f \in \{0, 1, \cdots, n-1\}^2$ is, at every point $(x, y) \in \{0, \cdots, n-1\}^2$,

$$f(x, y) = \sum_{k_1, k_2 = 0}^{n-1} fw(k_1, k_2) \times w_{k_1}(x) \times w_{k_2}(y) = \sum_{k_1, k_2} \left\langle f, w_{k_1} \otimes w_{k_2} \right\rangle (w_{k_1} \otimes w_{k_2})(x, y)$$

where, for every $(k_1, k_2) \in \{0, \cdots, n-1\}^2$,

$$fw(k_1, k_2) = \left\langle f, w_{k_1} \otimes w_{k_2} \right\rangle = \frac{1}{n^2} \sum_{x=0}^{n-1} \sum_{y=0}^{n-1} f(x, y) \times w_{k_1}(x) \times w_{k_2}(y)$$

is the (k_1, k_2) Haar coefficient of f.

In Figure 9, one observes that the great majority of the wavelet coefficients of an image are negligible compared with the few thousand largest coefficients.

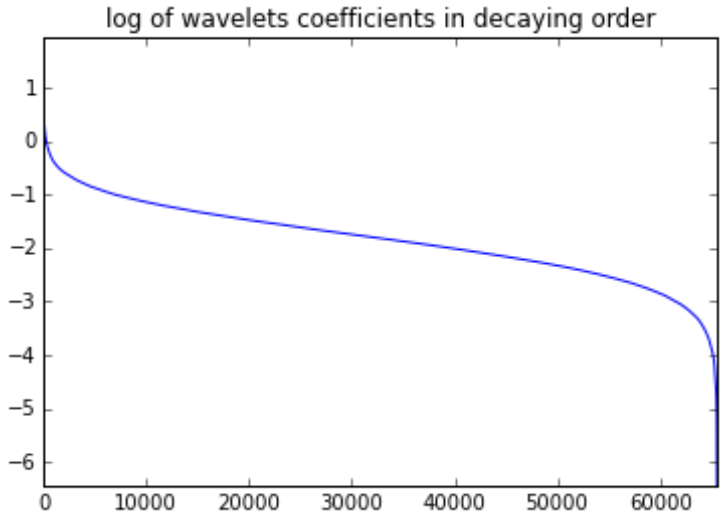


Figure 9: Logarithm of the wavelet coefficients of an image, ordered decreasingly. The great majority of the coefficients are very small compared with the first 4,000.

One may then threshold the smallest coefficients to zero and keep only the "handful" of largest ones. This is what is done in Figure 10.

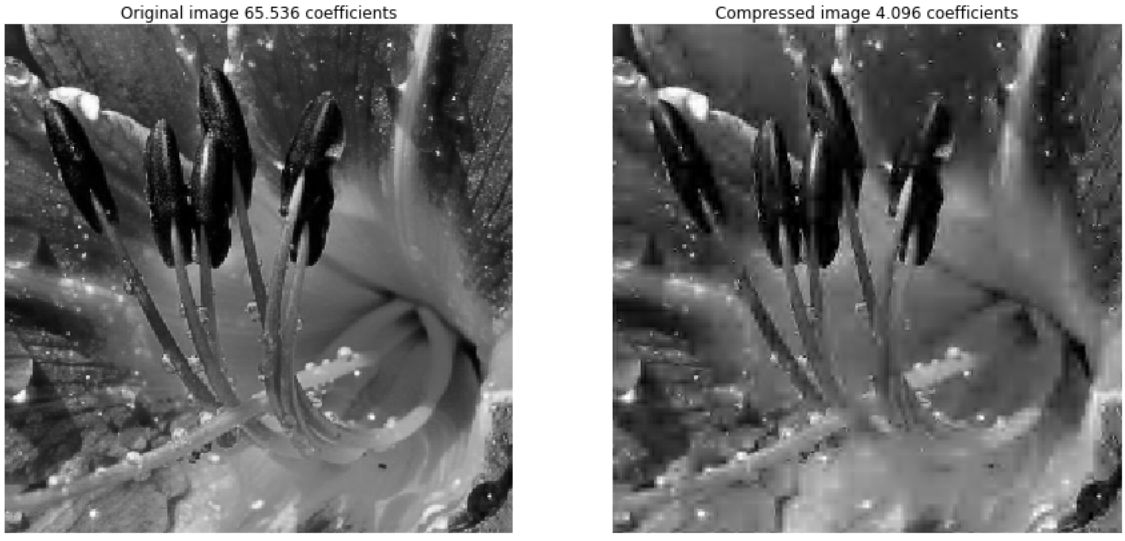


Figure 10: On the left, the original image. On the right, the image obtained by hard thresholding (only 16% of the coefficients are kept).

The sparsity assumption therefore seems to be satisfied. The construction of the single-pixel camera has produced the reconstruction results given in Figure 11.



Figure 11: Example of image reconstruction from measurements of the single-pixel camera.

4 The Compressed Sensing problem for partial Fourier measurements – the notion of incoherent measurements

Given a signal $x \in \mathbb{C}^N$ and a set $I \subset \{1, \dots, N\}$ of frequencies, it is in general impossible to reconstruct x from the Fourier coefficients $(\widehat{x}_i : i \in I)$ alone, unless $I = \{1, \dots, N\}$ (because the Fourier transform is a bijection of $\mathbb{C}^N$).

Nevertheless, if x is s-sparse, then one falls back into the framework of Compressed Sensing: reconstruct a structured signal (here, sparse in the canonical basis) from "incomplete information" on this signal (here, a small number of its Fourier coefficients).

To fix the notation, we introduce the Fourier matrix (the discrete Fourier transform):

$$\Gamma : \left\{ \begin{array}{ccc} \mathbb{C}^N & \to & \mathbb{C}^N \\ x & \to & \Gamma x = \widehat{x} \end{array} \right. \quad \text{where } \Gamma = \Big(\frac{w^{(p-1)(q-1)}}{\sqrt{N}} \Big)_{1 \leq p,q \leq N} \text{ and } w = \exp(-2\mathbf{i}\pi/N). \tag{4.1}$$

We denote by $\bar{\Gamma}_1, \dots, \bar{\Gamma}_N$ the row vectors of Γ. Thus $\widehat{x}_i = \langle \Gamma_i, x \rangle = \bar{\Gamma}_i^\top x$ for every $i = 1, \dots, N$. In particular, the observations $\widehat{x}_i$, $i \in I$, are indeed linear measurements of the signal x.

For a subset $I \subset \{1, \dots, N\}$ of frequencies, one has the data $(\widehat{x}_i : i \in I)$; in matrix terms, one has the datum $\Gamma_I x$, where

$$\Gamma_I : \left\{ \begin{array}{ccc} \mathbb{C}^N & \to & \mathbb{C}^I \\ x & \to & (\widehat{x}_i : i \in I). \end{array} \right.$$

This is therefore the measurement matrix introduced in (1.3), in which the measurement vectors are the $\bar{\Gamma}_i$ for $i \in I$—the number of measurements is therefore $m = |I|$ here. In this case, the measurement matrix is obtained by extracting the rows with indices in I from the (square) Fourier matrix $\Gamma \in \mathbb{C}^{N \times N}$.

The mathematical problem is stated here as follows: given that one observes $m = |I|$ Fourier coefficients of a signal x living in a high-dimensional space of dimension N, but whose support (in the canonical basis) has small size s, what are the conditions on the three parameters s, m, and N, and which sets $I \subset \{1, \dots, N\}$ of frequencies should be transmitted, so that it is possible to reconstruct exactly every s-sparse signal x from these m Fourier coefficients $\Gamma_I x$ alone?

We shall show later in the course that a random selection of the frequencies to be transmitted makes it possible to take m (the number of measurements) of the order of s (the size of the supports of the signals), up to logarithmic terms.

4.1 Introduction to the notion of incoherent measurements

The Fourier matrix Γ may here be replaced by any matrix Γ such that:

1. Γ is an isometry: $\forall x \in \mathbb{C}^N, \|\Gamma x\|_2 = \|x\|_2$,
2. the entries of Γ are such that for every $1 \leq p, q \leq N$, $|\Gamma_{pq}| \leq C/\sqrt{N}$ (where C is an absolute constant). That is, if $\bar{\Gamma}_1, \ldots, \bar{\Gamma}_N$ are the row vectors of Γ and $(e_1, \ldots, e_N)$ is the canonical basis of $\mathbb{C}^N$, then for every $1 \leq p, q \leq N$,

$$|\langle \Gamma_p, e_q \rangle| \leq \frac{C}{\sqrt{N}}. \tag{4.2}$$

Property (4.2) turns out to be a key point in the choice of the measurements. When property (4.2) is satisfied, one says that the orthonormal bases $(\bar{\Gamma}_1, \ldots, \bar{\Gamma}_N)$ and $(e_1, \ldots, e_N)$ are incoherent. Geometrically, this means that one basis is made up of diagonal elements of the other (and conversely). This idea appears constantly in CS in the choice of the measurement vectors: it is "necessary" that the measurement vectors be incoherent with the basis in which sparsity is expressed. Here sparsity is expressed in the canonical basis, so one seeks measurements that are incoherent with the canonical basis. If sparsity had been expressed in the Fourier basis, for example, then one would have taken measurement vectors from the canonical basis, for example (or from any other basis incoherent with the Fourier basis). We shall see later that a strong point of random measurements is that, given a basis, with high probability the random vector is incoherent with the elements of that basis. Of course, not every random vector satisfies this property (for example, Dirac vectors in the canonical basis), but for the typical case of a Gaussian vector (i.e., with i.i.d. centered reduced Gaussian entries), one observes this phenomenon, as well as for other examples of random vectors.

The notion of incoherence extends the duality between time and frequency; a signal that is highly localized in time will be highly spread out in frequency (this is the idea of the uncertainty principle, which will be discussed in more detail later in the course). Here an object that is sparse (in signal-processing terminology, one says *localized* in time) in a basis or a dictionary $\{f_1, \ldots, f_M\}$ must be well spread out (which is the opposite of sparse) in the dictionary formed by the measurements $\{X_1, \ldots, X_m\}$. In particular, the measurement vectors $\{X_1, \ldots, X_m\}$ have a well-spread representation in the basis in which sparsity is expressed.

One may define this notion of incoherence formally. Suppose that we have two bases $\mathcal{B}_f := (f_1, \ldots, f_N)$ and $\mathcal{B}_\Gamma := (\Gamma_1, \ldots, \Gamma_N)$ of $\mathbb{R}^N$. In the basis $\mathcal{B}_f$, the signals x that we wish to reconstruct have a sparse representation; $\mathcal{B}_f$ is therefore a good choice of basis for representing these signals. We shall call it the representation basis. The second basis $\mathcal{B}_\Gamma$ is used for taking measurements; that is, one will choose the measurement vectors in $\mathcal{B}_\Gamma$, but since one wishes to take only a small number of measurements, one chooses only m among the N vectors in $\mathcal{B}_\Gamma$. The basis $\mathcal{B}_\Gamma$ is called the measurement basis.

Remark 4.1 *If $\mathcal{B}_f$ is an orthonormal basis of $\mathbb{R}^N$ and $\mathcal{B}_0 = (e_1, \ldots, e_N)$ is the canonical basis, then every signal x is written in $\mathcal{B}_f$ and in $\mathcal{B}_0$ as*

$$x = \sum_{j=1}^{N} \langle x, f_j \rangle f_j = \sum_{j=1}^{N} x_j e_j.$$

One then says that $(x_j)_{j=1}^N$ is the representation of x in the time domain and that $(\langle x, f_j \rangle)_{j=1}^N$ is the representation of x in the $\mathcal{B}_f$-domain. This is the terminology used in signal processing, which

extends the jargon introduced for Fourier analysis: when $\mathcal{B}_f$ is the Fourier basis (in that case, the space of signals is $\mathbb{C}^N$ rather than $\mathbb{R}^N$), the $\mathcal{B}_f$-domain is better known as the frequency domain, or the Fourier domain.

Definition 4.2 *Let $(\mathcal{B}_f, \mathcal{B}_\Gamma)$ be a pair of orthonormal bases of $\mathbb{R}^N$. The* ***coherence*** *between the representation basis $\mathcal{B}_f$ and the measurement basis $\mathcal{B}_\Gamma$ is*

$$\mu(\mathcal{B}_f, \mathcal{B}_\Gamma) = \max_{1 \leq p,q \leq N} |\langle \Gamma_p, f_q \rangle|.$$

One always has $\mu(\mathcal{B}_f, \mathcal{B}_\Gamma) \in [1/\sqrt{N}, 1]$. Indeed, for every p, q, $|\langle \Gamma_p, f_q \rangle| \leq \|\Gamma_p\|_2 \|f_q\|_2 = 1$, and by Parseval's identity, for every p, $\sum_q |\langle \Gamma_p, f_q \rangle|^2 = 1$, so necessarily at least one term $|\langle \Gamma_p, f_q \rangle|^2$ in this sum is larger than $1/N$.

In CS, one is mainly interested in pairs of representation and measurement bases with small coherence, of order $1/\sqrt{N}$. This is in particular the case of the pair (canonical basis, Fourier basis), and it is one of the reasons why the reconstruction of s-sparse vectors in the canonical basis can be carried out from only a small number of their Fourier coefficients.

In the opposite case of bases with large coherence—for example, reconstructing s-sparse vectors in the canonical basis from measurements obtained by also choosing measurement vectors in the canonical basis—the approach is doomed to fail, because one will need $m = N$ measurements, whereas for the same problem only $m \sim s$ Fourier measurements are sufficient. We refer the interested reader to the article [2] for theoretical results relating the number of measurements to the coherence between the representation basis and the measurement basis.

5 Compressed sensing for radar data

Many radar data can be interpreted as linear measurements of a given scene. In some cases one may then use the ideas developed in CS to process these data and to get an idea of which signals to emit in order to gain in precision and in transmission time.

In this section, we consider the setting of a ground-based radar whose purpose is to identify aircraft in flight within its surveillance perimeter. The objective of the radar is therefore to identify the position and the velocities of these aircraft. To this end, the radar scans its surveillance space and emits signals in order to determine the distance and the speed of potential targets in its field of surveillance. Thus, from the continuous observation of the distance and the speed of these targets, one can determine their locations and their velocities.

At each emission of a signal, the radar must determine the position and the speed of each of the targets present. It turns out that the distance of a target can be obtained from the presence of a shift in the emitted wave, and the speed is obtained by the Doppler effect, which induces a frequency modulation of the emitted wave. A target can therefore be located (its position and its speed) by a time shift and a frequency modulation: knowing this time shift and this frequency modulation, one can determine the distance (to the radar) and the speed of the target.

A scene may then be described by a few targets, each inducing a time shift and a frequency modulation. One discretizes the shift and the modulation (which will lead to solving an interpolation problem in post-processing when the "true" shifts and modulations do not lie on the discrete grid—we shall not treat this problem here).

One then introduces two operators $T_k : \mathbb{C}^N \mapsto \mathbb{C}^N$ and $M_l : \mathbb{C}^N \mapsto \mathbb{C}^N$ for every $0 \leq k, l \leq N-1$, such that for every $z \in \mathbb{C}^N$ and every $0 \leq j \leq N-1$,

$$(T_k z)_j = z_{j-k \bmod N} \text{ and } (M_l z)_j = \exp\Big(\frac{2\mathbf{i}\pi}{N} jl\Big) z_j \tag{5.1}$$

where " mod N" means *modulo* N. The operator T_k is a time-shift operator; it performs a *shift* of length k. The operator M_l performs a frequency modulation due to the Doppler effect of a moving target.

A scene may therefore be modeled by a signal of the form

$$x = \sum_{0 \leq k,l \leq N-1} x_{k,l} T_k M_l \in \mathbb{C}^{N \times N} \tag{5.2}$$

where the nonzero entries of the coefficient vector $(x_{k,l})_{0 \leq k,l \leq N-1}$ give the time shifts and the frequency modulations of the targets present (from which one can deduce their distances to the radar and their speeds).

Sparsity comes from the fact that the airspace is clear most of the time and that at most about ten targets are present. The signal x expressed in the dictionary $\mathcal{F} := \{T_k M_l : 0 \leq k, l \leq N-1\}$ is therefore sparse. One may write a matrix representation of the elements of this dictionary:

$$T_k M_l = \begin{pmatrix} 0 & 0 & \cdots & 0 & \exp\left(\frac{2\mathbf{i}\pi lk}{N}\right) & 0 & 0 & \cdots \\ 0 & 0 & \cdots & 0 & 0 & \exp\left(\frac{2\mathbf{i}\pi l(k+1)}{N}\right) & 0 & \cdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \ddots & \cdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & \exp\left(\frac{2\mathbf{i}\pi l(N-1)}{N}\right) \\ 1 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots \\ 0 & \exp\left(\frac{2\mathbf{i}\pi l}{N}\right) & \cdots & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & \ddots & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & \cdots & \exp\left(\frac{2\mathbf{i}\pi l(k-1)}{N}\right) & 0 & 0 & 0 & \cdots \end{pmatrix}$$

from the expression, for every $0 \leq p, q \leq N-1$,

$$(T_k M_l)_{p,q} = \begin{cases} 0 & \text{if } p + k \bmod N \neq q \\ \exp\left(\frac{2\mathbf{i}\pi lq}{N}\right) & \text{if } p + k \bmod N = q. \end{cases}$$

Thus the signal x that we wish to estimate here is an $N \times N$ matrix that is sparse (when it is expressed in the dictionary $\mathcal{F}$). The natural approach in CS would then be to construct "measurement matrices" of size $N \times N$ and to observe the inner product between the signal x and these matrices. Unfortunately, this has no physical meaning. Here, as in many other problems, the choice of the measurements is constrained by the problem (and even in some problems—cf. Section 7—one has no choice of the measurements).

In the case of radar signals, the observations (which are naturally corrupted by noise from the environment and from the components of the radar itself, but which, to simplify the exposition, are treated as noiseless) are of the form

$$y = xg \tag{5.3}$$

where $g \in \mathbb{C}^N$ is a vector made up of the pulses emitted by the radar. This vector is left to the free choice of the radar designer (within the physical limits of the radar). The choice of g must be made so that x can be reconstructed from $y = xg$. Here one has no choice about the number of measurements, which is necessarily N, whereas the dimension of x is N^2. A popular choice is the *Alltop* vector $g \in \mathbb{C}^N$ such that

$$g_l = \exp\left(\frac{2\mathbf{i}\pi}{N} l^3\right) \text{ for all } l = 0, \ldots, N-1.$$

Remark 5.1 *The radar example, like the examples that follow, highlights a fairly classical situation in CS concerning the choice of the measurement vectors. In the modeling proposed in Section 2, the statistician chooses the measurement vectors—the notion of incoherence between measurement vectors and the representation basis can guide that choice (and the subsequent lectures will give other ideas on the choice of these measurement vectors)—but in the radar case the choice of these measurement vectors is very limited; here even the number of measurements is imposed by the physics of the problem. In this case, one can only hope that these measurements are "naturally" suited to the CS problem, and in particular that these measurements are sufficiently incoherent with the representation basis.*

Following Remark 5.1, one must check that the measurements obtained for this radar detection problem have a chance of being incoherent with the representation basis. To this end, we rewrite the observed vector y as a linear measurement of the sparse vector $X = (x_{k,l})_{0\leq k,l\leq N-1}$ (which is the representation of x in $\mathcal{F}$) by

$$y = xg = A_g X \text{ where } A_g \in \mathbb{C}^{N\times N^2}$$

and the N^2 columns of A_g are the vectors $T_k M_l g \in \mathbb{C}^N$ for every $0 \leq k, l \leq N-1$. In particular, if g is of the form

$$g = \begin{pmatrix} \exp\left(\frac{2\mathbf{i}\pi p_0 \times 0}{N}\right) \\ \exp\left(\frac{2\mathbf{i}\pi p_1 \times 1}{N}\right) \\ \vdots \\ \exp\left(\frac{2\mathbf{i}\pi p_j \times j}{N}\right) \\ \vdots \\ \exp\left(\frac{2\mathbf{i}\pi p_{N-1} \times (N-1)}{N}\right) \end{pmatrix} = \left(\exp\left(\frac{2\mathbf{i}\pi p_l \times l}{N}\right)\right)_{l=0}^{N-1}$$

where $p_0, p_1, \ldots, p_{N-1} \in \{0, 1, \ldots, N-1\}$ are frequencies emitted by the radar (left to the operator's choice).

6 Compressed sensing for face recognition

In this section we present the face-recognition problem, which we shall write as a CS problem. The approach used here is called *sparse representation-based classification (SRC)*. A detailed exposition of this approach can be found in Chapter 12 of [3].

We are given a data set of the form $\mathcal{D} := \{(\phi_j, \ell_j) : 1 \leq j \leq N\}$, where:

1. $\phi_j \in \mathbb{R}^m$ is a vector representation of the j-th image; for example, one may think of ϕ_j as the (concatenated) vector of the pixels of an image of size $p \times q$, and hence $m = pq$,

2. $\ell_j \in \{1, \ldots, C\}$ is a label indicating the index of the individual represented in the j-th image.

The same individual may be represented several times, from different angles, under different lighting conditions, and so on, in the database $\mathcal{D}$.

We are given a new image $y \in \mathbb{R}^m$, and we wish to assign to y (automatically, from the data in $\mathcal{D}$) a label in $\{1, \ldots, C\}$ indicating the index of the individual represented in the new image y. This is the face-recognition problem, which may be viewed as a multiclass classification problem.

In what follows, we rewrite this problem in the CS framework. Results (cf. [1, 5]) that have had a great influence in the face-recognition community have observed that if the number of images of a given subject is large enough, and if there is sufficient diversity among these photographs (in terms of exposure to light and of camera angles), then these images lie close to a low-dimensional linear subspace of the high-dimensional space of dimension m. For a certain representation ϕ of the images, this dimension may even be taken equal to 9 (cf. [1]). Thus, if the data set $\mathcal{D}$ is sufficiently rich in size and varied in exposure and camera angles, then one may hope that the new image y of a subject $i \in \{1, \dots, C\}$ (for some i to be determined) will be of the form

$$y \approx \sum_{j:\ell_j=i} \phi_j x_j. \tag{6.1}$$

We therefore assume that the new image (represented by y) is a linear combination of the images of subject i already present in the data set $\mathcal{D}$. One may rewrite this assumption in the CS formalism:

$$y = \Phi x + \xi \tag{6.2}$$

where ξ is viewed as a noise term (due to the linear approximation of y by the space spanned by $\{\phi_j : \ell_j = i\}$) and

1. $\Phi := [\Phi_1|\Phi_2|\cdots|\Phi_C]$, where $\Phi_i = [\phi_j : \ell_j = i]$ for every $i \in \{1, \dots, C\}$. That is, one concatenates the images of individual i present in the database $\mathcal{D}$ into the matrix Φ_i—in particular, if there are n_i photographs of individual i in $\mathcal{D}$, then Φ_i has size $m \times n_i$. One then concatenates the Φ_i for $i = 1, \dots, C$ into the matrix Φ, which therefore has size $m \times N$.

2. the vector $x \in \mathbb{R}^N$ has the block form

$$x = [\mathbf{0}^\top|\mathbf{0}^\top|\cdots|\mathbf{x}_i^\top|\mathbf{0}^\top|\cdots|\mathbf{0}^\top]^\top$$

   where $\mathbf{x}_i$ is the restriction of x to the block of columns associated with Φ_i in Φ.

One then sees that the face-recognition problem can be cast in the form of the noisy CS model, since the signal x to be recovered is sparse in the canonical basis. Here the measurement vectors are the rows of Φ, which are formed from the representations of the photographs in the database.

The problem we mainly wish to solve here is the reconstruction of the support of x, which makes it possible to identify the individual represented in y: if one constructs an estimator $\widehat{x}$ of x whose support is mainly contained in the block associated with Φ_i in Φ, then one may conclude that y represents individual i.

The construction of the measurement matrix therefore depends on the observations contained in the database $\mathcal{D}$. Either this database is provided in advance and one has no way to influence its construction. Or one may construct it oneself, as in Figures 12 and Figure 13. In the latter case, one may expose each individual to different light levels from different angles; this is how the database (and hence the measurement matrix) is constructed.



Figure 12: For the face-recognition problem, constructing the database is equivalent to constructing the measurement matrix in CS: here, photographs taken from different angles.



Figure 13: For the face-recognition problem, constructing the database is equivalent to constructing the measurement matrix in CS: here, photographs taken under different lighting conditions.

7 Compressed sensing for financial data

Imagine that we have access, in real time, to the value of a portfolio of stocks (or of other financial assets); we denote these values, taken for example every minute, by $y_1, y_2, \cdots$. We also assume that this portfolio is not very active and that over one week its structure remains the same. Finally, thanks to Bloomberg and Reuters data, one has access to the instantaneous prices of the stocks on the market in which this portfolio operates (cf. Figure 14).

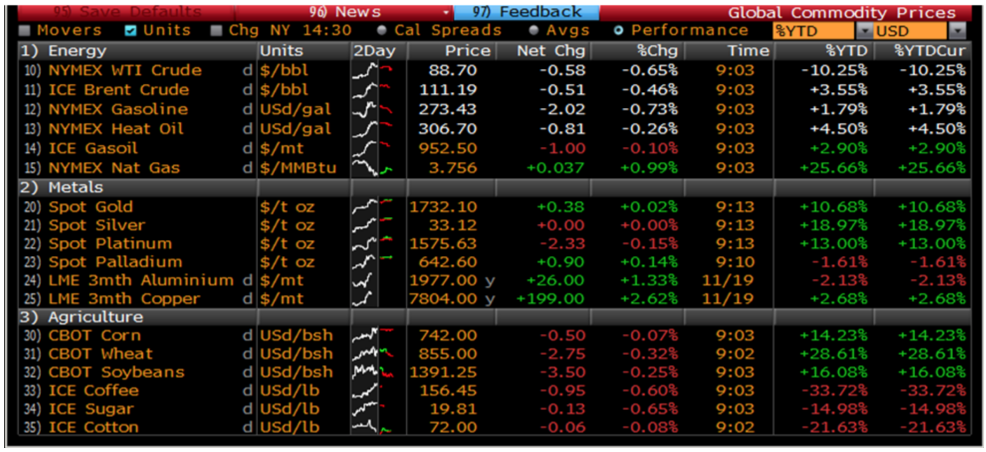


Figure 14: Financial data available on the *Bloomberg* website. Here the measurement matrix is built from the values taken by the stocks over time.

Stock j has price $x_{i,j}$ at time i. The data available to us are therefore of the form:

$$\begin{aligned} t=1: &\quad y_1: \text{ portfolio value} \quad (x_{1,j})_{j=1}^N: \text{ stock prices} \\ t=2: &\quad y_2: \text{ portfolio value} \quad (x_{2,j})_{j=1}^N: \text{ stock prices} \\ &\cdots\cdots \\ t=m: &\quad y_m: \text{ portfolio value} \quad (x_{m,j})_{j=1}^N: \text{ stock prices} \end{aligned}$$

We seek to understand how this portfolio is structured; that is, to identify which assets make it up and in what quantity.

One may pose this problem in the CS framework. We denote by $y = (y_j)_{j=1}^m$ the values taken by the portfolio and by $A = (x_{i,j} : 1 \le i \le m, 1 \le j \le N)$ the matrix of prices. We then wish to recover an $x \in \mathbb{R}^N$ such that $y = Ax$.

The idea that the stock portfolio in fact contains only very few distinct stocks translates directly into a sparsity assumption on x: one expects x to have a small support.

Reconstructing the support of x and estimating the values taken by x on its support will therefore solve the problem we wish to treat here.

8 Notation

Finally, in this last section, we collect the list of notation that we shall use throughout this course.

For every $p > 0$, one defines the norms (when $p \ge 1$) and quasi-norms (when $0 < p < 1$), ℓ_p^d on $\mathbb{R}^d$, by

$$\|x\|_p = \Big(\sum_{j=1}^d |x_j|^p \Big)^{1/p}$$

for every $x \in \mathbb{R}^d$. The space $\mathbb{R}^d$ equipped with the norm $\|\cdot\|_p$ is denoted ℓ_p^d. The norm $\|\cdot\|_p$ will sometimes be called the ℓ_p norm, or the ℓ_p^d norm. One then identifies the norm and the space equipped with that norm when there is no ambiguity. The unit ball of the space ℓ_p^d is

$$B_p^d = \big\{ x \in \mathbb{R}^d : \|x\|_p \le 1 \big\}$$

and the unit sphere of ℓ_p^d is denoted

$$S_p^{d-1} = \big\{ x \in \mathbb{R}^d : \|x\|_p = 1 \big\}.$$

For every $I \subset \{1, \dots, N\}$, we denote by $P_I x$ the vector in $\mathbb{C}^N$ that coincides with x on the coordinates in I and that vanishes on the complement of I. For every $T \subset \mathbb{C}^N$, $P_I T = \{P_I t : t \in T\}$.

The set of all s-sparse vectors in $\mathbb{R}^N$ is denoted

$$\Sigma_s = \{x \in \mathbb{R}^N : \|x\|_0 \le s\}.$$

If $A \in \mathbb{R}^{m \times N}$, and $I \subset \{1, \dots, m\}$, $J \subset \{1, \dots, N\}$, then

1. $A_{I\cdot}$ is the matrix extracted from A in which only the rows with indices in I have been kept,
2. $A_{\cdot J}$ is the matrix extracted from A in which only the columns with indices in J have been kept,
3. A_{IJ} is the matrix extracted from A in which only the rows with indices in I and the columns with indices in J have been kept.

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